thermore, adding a wireless connection to smoke detectors enables additional features that increase safety and convenience.

Pressure sensors. These sensors are used in IoT systems to monitor systems and devices that are driven by pressure signals. When the pressure range is beyond the threshold level, the device alerts the user about the problems that should be fixed. For example, BMP180 is a popular digital pressure sensor for use in mobile phones, PDAs, GPS navigation devices and outdoor equipment. Pressure sensors are also used in smart vehicles and aircrafts to determine force and altitude, respectively. In vehicle, tyre pressure monitoring system (TPMS) is used to alert the driver when tyre pressure is too low and could create unsafe driving conditions.

Fig. 2: Temperature sensors

Fig. 3: HPP801A031 humidity sensor

Fig. 4: PIR motion sensor

Image sensors. These sensors are found in digital cameras, medical imaging systems, night-vision equipment, thermal imaging devices, radars, sonars, media house and biometric systems. In the retail industry, these sensors are used to monitor customers visiting the store through IoT network. In offices and corporate buildings, they are used to monitor employees and various activities through IoT networks.

Accelerometer sensors. These sensors are used in smartphones, vehicles, aircrafts and other applications to detect orientation of an object, shake, tap, tilt, motion, positioning, shock or vibration. Different types of accelerometers include Hall-effect accelerometers, capacitive accelerometers and piezoelectric accelerometers.

IR sensors. These sensors can measure the heat emitted by objects. They are used in various IoT projects including healthcare to monitor blood flow and blood pressure, smartphones to use as remote control and other functions, wearable devices to detect amount of light, thermometers to monitor temperature and blind-spot detection in vehicles.

Proximity sensors. These sensors detect the presence or absence of a nearby object without any physical contact. Different types of proximity sensors are inductive, capacitive,